

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0114

Winthrop 230 kV
March 28, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0114. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Winthrop 230 kV substation in the Ohio area. The request was received on March 09, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 98 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$67,522,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Truman - Lookingglass 230 kV	Swan Vly - Trego 500 kV	125.4	398	10.1
Lookingglass - Corridor 230 kV	Swan Valley - Trego 500 kV	125.2	423	10.1
Corridor - Larson St 230 kV line	Swan Vly - Trego Substation 500 kV	119.7	465	10.1
Minco - Larson St 230 kV	Barrett - Norwich 500 kV line	119.1	498	33.5
Waldo - Truman 230 kV Line	Swan Vly - Trego 500 kV	108.9	436	10.1
Old Forge - Portage Lakes 230 kV	Surrowiec - Weedpatch 500 kV L	105.1	574	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Truman - Lookingglass 230 kV	Rebuild line with higher-capacity conductor	\$17,262,000	80% (\$13,728,000)
Lookingglass - Corridor 230 kV Line	Rebuild line with higher-capacity conductor	\$23,658,000	80% (\$18,814,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Corridor - Larson St 230 kV line	Rebuild line with higher-capacity conductor	\$19,406,000	80% (\$15,433,000)
Minco - Larson St 230 kV	Rebuild line with higher-capacity conductor	\$8,963,000	88% (\$7,909,000)
Waldo - Truman 230 kV Line	Reconductor line	\$9,864,000	80% (\$7,845,000)
Old Forge - Portage Lakes 230 kV	Reconductor line	\$6,101,000	62% (\$3,793,000)
	Total allocated to project		\$67,522,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.